INDIVIDUAL REFLECTION

Python Application Development — Industry-Oriented Assignment

1. Design and Development Decisions

This assignment helped me understand how individual Python concepts can be integrated into a complete, practical application rather than being used only in separate classroom programs. One of the key design decisions was selecting appropriate data structures. Customer and food-item records require quick retrieval using unique identifiers, so dictionaries were more suitable than repeatedly searching through lists. Orders were stored as a list of dictionaries because each order contains multiple related fields and the collection needs to be filtered and processed for reports. This design connected Python concepts directly with CO3.

2. Challenges Faced

A major challenge was ensuring that the program could handle incorrect data without stopping execution unexpectedly. Situations such as duplicate identifiers, missing records, invalid order statuses and unavailable CSV files had to be considered. I learned to place validation close to the operation where it is required and to use a custom exception for application-specific errors. Control-flow statements and loops were also essential for searching, filtering and aggregating order information. These activities strengthened my understanding of CO1 and CO2 by combining functions, expressions, string methods, conditions and iteration in one application.

3. Learning Gained

Testing was another important learning experience. Instead of assuming that a function was correct simply because it executed successfully, I learned to compare expected results with actual output. For example, an order containing two items priced at Rs.80 each should produce a total of Rs.160. Similarly, the cancellation percentage should be calculated using the number of cancelled orders divided by the total number of orders for that item. This demonstrated that proper validation involves checking both successful execution and the accuracy of the results.

4. SDG Relevance

The assignment is also connected to SDG 8 — Decent Work and Economic Growth. A simple digital ordering system can reduce repetitive manual record-keeping and help food-service teams use information more efficiently. At the same time, automation should protect customer privacy, maintain accurate billing information and remain simple enough for the intended users to understand and operate.

5. Contribution to Mapped Course Outcomes (COs)

Overall, this assignment contributed to the mapped course outcomes by transforming Python fundamentals, control flow and data structures into a practical and testable solution. It improved my ability to design programs, validate input, process data and test outputs systematically. If I continue developing the project, I would replace CSV storage with a database, add a user-friendly interface, implement stronger authentication and introduce more comprehensive automated testing.